

MADRID-BARAJAS AP, Spain

WMO: 082210

Lat: 40.467N

Lon: 3.556W

Elev: 1909

StdP: 13.71

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.7	28.1	14.0	12.0	39.0	18.6	15.0	40.1	26.1	49.9	23.1	48.5	3.6	350	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.0	98.2	65.1	95.9	64.5	93.4	63.8	69.1	90.5	67.1	88.5	65.6	87.3	9.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.5	87.6	75.7	59.8	82.4	73.7	58.3	77.9	72.3	34.2	90.6	32.6	88.3	31.4	87.4	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	21.6	18.7	16.5	DB	20.9	102.3	3.2	2.2	18.6	103.9	16.7	105.2	14.9	106.5	12.5	108.1
(5)				WB	19.6	70.5	3.3	3.1	17.2	72.8	15.2	74.6	13.3	76.3	10.9	78.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.0	43.0	45.1	50.8	55.2	62.6	72.5	78.1	77.5	69.2	60.2	49.4	43.6	(6)
(7)		DBStd	13.63	5.04	4.88	5.17	5.74	6.54	6.37	5.00	4.49	5.51	5.35	5.64	5.27	(7)
(8)		HDD50	733	223	148	54	17	2	0	0	0	3	78	208		(8)
(9)		HDD65	3436	683	557	441	297	123	12	1	0	23	165	469	665	(9)
(10)		CDD50	4020	6	10	78	172	392	676	873	851	575	318	60	9	(10)
(11)		CDD65	1245	0	0	0	2	48	238	408	386	148	15	0	0	(11)
(12)		CDH74	16831	0	0	7	119	774	3266	5644	5032	1749	240	0	0	(12)
(13)		CDH80	8369	0	0	0	9	239	1600	3146	2681	652	42	0	0	(13)
(14)	Wind	WSAvg	6.4	6.0	6.8	7.1	7.4	6.6	6.8	6.4	5.9	5.6	6.0	5.4	(14)	
(15)	Precipitation	PrecAvg	15.1	1.2	1.2	1.2	1.6	1.7	0.8	0.3	0.3	0.9	2.4	1.9	1.5	(15)
(16)		PrecMax	21.5	3.7	3.7	3.7	3.7	4.1	2.4	1.3	1.1	2.6	6.2	7.2	5.0	(16)
(17)		PrecMin	9.4	0.0	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.3	0.1	0.1	(17)
(18)		PrecStd	3.1	1.0	0.8	0.9	0.8	1.2	0.7	0.3	0.4	0.6	1.5	1.5	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	61.2	67.1	74.9	81.1	90.9	100.4	101.6	100.3	95.0	84.9	71.0	60.9	(19)	
(20)		MCWB	50.2	51.0	54.4	58.7	61.7	65.3	66.1	65.0	64.6	60.3	56.1	51.7	(20)	
(21)		DB	57.5	62.8	71.4	77.6	86.3	95.6	98.5	97.3	90.0	80.2	66.4	57.5	(21)	
(22)		MCWB	49.0	49.6	53.0	57.0	60.4	64.4	65.5	64.7	62.6	59.2	54.8	50.6	(22)	
(23)		DB	54.9	59.2	68.0	73.7	82.1	92.4	96.5	95.1	86.8	76.2	62.7	55.3	(23)	
(24)		MCWB	47.9	48.5	52.0	55.9	59.2	63.2	64.9	64.4	61.6	58.4	53.0	49.5	(24)	
(25)		DB	52.1	55.9	64.4	69.5	78.3	89.2	93.8	92.5	83.8	72.4	59.3	52.9	(25)	
(26)		MCWB	46.4	47.2	50.6	53.6	58.0	62.4	63.8	63.7	60.9	57.4	51.8	47.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	53.7	54.5	57.1	60.5	64.8	69.8	72.6	71.3	67.5	64.1	59.3	55.3	(27)	
(28)		MCDWB	58.1	61.3	69.6	76.7	83.3	91.3	95.9	92.8	85.5	76.6	66.1	57.8	(28)	
(29)		WB	51.1	51.9	54.9	58.5	62.4	66.9	69.4	68.9	65.7	62.1	56.5	52.5	(29)	
(30)		MCDWB	55.2	59.3	67.5	74.7	80.5	88.6	92.0	89.9	83.6	72.8	63.1	55.9	(30)	
(31)		WB	49.1	49.8	53.3	56.6	60.7	65.3	67.0	67.0	64.1	60.7	54.7	50.3	(31)	
(32)		MCDWB	53.2	57.2	64.6	71.1	77.6	87.2	90.3	87.8	81.0	70.4	60.6	54.0	(32)	
(33)		WB	47.3	48.1	51.5	54.7	59.2	63.8	65.3	65.3	62.6	59.2	52.9	48.5	(33)	
(34)		MCDWB	51.4	54.7	61.7	66.8	74.4	84.7	88.7	86.8	78.4	68.5	58.0	52.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	17.2	20.0	22.4	21.8	23.5	26.9	28.0	27.5	24.9	21.2	17.6	16.9	(35)	
(36)		5% DB	22.7	27.5	31.6	30.6	31.5	31.8	30.8	31.0	30.2	28.8	23.4	20.5	(36)	
(37)		MCWBR	14.8	16.1	16.6	14.2	12.0	10.6	11.0	11.5	11.7	12.9	13.4	13.8	(37)	
(38)		5% WB	16.8	22.0	26.5	26.9	26.7	28.3	28.9	27.9	25.3	21.0	18.3	15.3	(38)	
(39)		MCWBR	12.1	13.9	14.7	13.1	11.1	10.8	12.3	12.2	10.8	10.6	11.4	11.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.306	0.341	0.350	0.364	0.367	0.349	0.359	0.348	0.334	0.306	0.291	(40)	
(41)		taud	2.508	2.478	2.398	2.381	2.368	2.380	2.403	2.381	2.423	2.477	2.512	2.517	(41)	
(42)		Ebn at Noon	279	290	289	293	290	287	292	286	283	275	270	270	(42)	
(43)		Edh at Noon	23	28	34	37	38	38	37	36	33	28	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	669	959	1350	1718	2058	2352	2455	2128	1637	1095	736	601	(44)	
(45)		RadStd	95	116	158	158	169	104	75	69	66	100	98	70	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+1.04	N/A	N/A	N/A	N/A	N/A	+137
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+1.28	N/A	N/A	N/A	N/A	N/A	+110

Nomenclature: See separate page